Game Theory Thirteen

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Contents

1 Candidacy Topics

- **Paper:** *From Mean Field Games to Navier–Stokes Equations* by Luo and Song.
- **Numerical Analysis:**
  - Initial value problems for ODEs (Chapter 5, LeVeque)
  - Absolute stability for ODEs (Chapter 7, LeVeque)
- **Stochastic Calculus:**
  - Brownian motion calculus (Chapter 4, Klebaner)
  - Stochastic differential equations (Chapter 5, Klebaner)
- **Fluid Mechanics and Hamiltonian Dynamics:**
  - Fluid mechanics (Chapters 1.1-1.11 and 2.1, Landau and Lifshitz)
  - Hamiltonian dynamics (Chapter 1, Swaters)

The only reference approval I am waiting on is the Swaters chapter, which will be resolved by Wednesday.

2 Standard ReLU with New Payoff Matrices

I started by considering a modified two-player, two-strategy game in which neither player had a strictly dominant strategy and the Nash Equilibrium is not located at $(\frac{1}{2}, \frac{1}{2})$

$$A = \begin{pmatrix} 2 & 4 \\ \\ 11 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 8 & 5 \\ \\ 6 & 23 \end{pmatrix}.$$

For these payoff matrices, the Nash equilibrium is

$$p^{1,*} = \left(\frac{17}{20}, \frac{3}{20}\right), \qquad p^{2,*} = \left(\frac{1}{4}, \frac{3}{4}\right).$$

Before testing Soft ReLU ($\varphi(x) = \ln(1 + e^x)$) with this game, I first ran some numerical experiments with the standard ReLU ($\varphi(x) = [x]_+ = \max\{0, x\}$). As expected, when the strategy trajectory crossed a threshold corresponding to a player's Nash Equilibrium, the error order of the BNN dynamics decreased from the expected fourth order to second order. Additionally, the BNN system converged to the Nash equilibrium in first order with respect to time.

When generating the error orders, I noticed that BNN dynamics was unstable with a time step of one, demonstrating that larger payoff magnitudes can make larger time steps unstable. To reduce improve numerical stability, I scaled the payoff matrices.

3 Scaling the Payoff Matrices

The payoff matrices were then linearly scaled so that the minimum payoff was zero and the maximum payoff was one. As Nash equilibria are invariant under linear transformations of payoff matrices, the fixed points of BNN dynamics remained unchanged.

$$A_{scaled} = \begin{pmatrix} \frac{1}{10} & \frac{3}{10} \\ \\ 1 & 0 \end{pmatrix}, \qquad B_{scaled} = \begin{pmatrix} \frac{1}{6} & 0 \\ \\ \frac{1}{18} & 1 \end{pmatrix}.$$

The numerical experiments suggested that smaller payoff magnitudes producee:

- lower numerical errors,
- less overflow risk,
- improved stability for larger step sizes,
- and slower but more controlled convergence.

4 Soft ReLU Activation Function

Having studied the new payoff matrices with ReLU, the next step was modeling BNN dynamics with a soft ReLU function for φ. This activation function performed excellently with the simple payoff matrices of last week. The question remained: what happens when the Nash equilibrium and payoff matrices are not simple?

This week's expeiments showed that Soft ReLU, while accurate at large step sizes, does not necessarily converge to the Nash equilibrium. The main issue is that BNN dynamics require the Nash equilibrium to be a fixed point of the dynamical system.

For the standard BNN dynamics,

$$\dot{p}_{ij} = \varphi_{ij(p)} - p_{ij} \sum_k \varphi_{ik}(p).$$

At a fixed point, we need

$$\varphi_{ij(p)} = p_{ij} \sum_k \varphi_{ik}(p).$$

This can happen one of two ways:

1. $\varphi_{ij(p)} = 0 \quad \forall i, j$
2. $p_{ij} = \frac{\varphi_{ij(p)}}{\sum_k \varphi_{ik}(p)} \quad \forall i, j$

As Soft ReLU is always positive, the fixed points of BNN dynamics with Soft ReLU must satisfy the second condition. This means that replacing ReLU with Soft ReLU changes the underlying equilibrium structure. With the exception of rare circumstances, the second condition is not satisfied at Nash equilibrium. While we could engineer φ to be zero at Nash equilibrium, doing so would require knowledge of the Nash equilibrium itself, defeating the purpose of BNN.

5 Recovering Nash Equilibria from Strategy Trajectories

If the Nash equilibrium is no longer a fixed point of the modified BNN system, an alternative approach is to ask whether the Nash equilibrium can be recovered from the trajectory.

One possible approach may be to average over the strategy trajectory after a certain period of time. This is conceptually similar to alternating fictitious play, where empirical averages of strategies converge to the Nash equilibrium. However, alternating fictitious play has additional assumptions, such as the existence of an ordinal potential (and thus a pure strategy Nash equilibrium).

6 Activation Function Engineering

One of my goals is to construct an activation function φ that improves numerical smoothness while preserving the important equilibrium properties of BNN dynamics.

The desired conditions are:

1. φ is increasing.
2. $\varphi(0) = 0$.
3. $\phi(x) = 0$ for $x \leq 0$, or is as close to zero as possible.
4. ϕ is smooth, or at least smoother than ReLU.

An open question is whether convexity of the activation function matters for convergence or stability.

7 Accuracy and Jacobian Switching

When RK4 is applied to a smooth ODE, fourth-order convergence is expected. However, BNN dynamics with ReLU are only piecewise smooth.

The Jacobian changes whenever an excess payoff crosses zero. These switching surfaces can reduce the observed convergence order. In previous experiments, the error order dropped from fourth order to approximately second order when trajectories crossed multiple switching regions.

This also explains why error plots can appear jagged. Changing the step size can change where/how the numerical trajectory crosses the switching surfaces.

8 Evolutionary Dynamics

The primary application of BNN dynamics these days seems to be in evolutionary population dynamics. In this framework, probability distributions are replaced with population distributions over strategies or attributes. The growth/decay of each population depends on the game governing interactions between individuals. Below are five methods of modeling population growth/decay from the textbook *Population Games and Evolutionary Dynamics*:

Revision protocol	Mean dynamic	Name (interpretation)
$\rho_{ij} = x_j[\pi_j - \pi_i]_+$	$\dot{x}_i = x_i \hat{F}_i(x)$	**Replicator** Growth is proportional to own share and excess payoff.
$\rho_i = M(\pi)$	$\dot{x} \in M(F(x)) - x$	**Best response** Population moves toward the set of best responses.
$\rho_{ij} = \dfrac{\exp(\eta^{-1}\pi_j)}{\sum_{k\in S}\exp(\eta^{-1}\pi_k)}$	$\dot{x}_i = \dfrac{\exp(\eta^{-1}F_i(x))}{\sum_{k\in S}\exp(\eta^{-1}F_k(x))} - x_i$	**Logit** Choice probabilities follow the logit (Gibbs) rule with precision η.
$\rho_{ij} = \left[\pi_j - \sum_{k\in S} x_k\pi_k\right]_+$	$\dot{x}_i = [\hat{F}_i(x)]_+ + x_i \sum_{j\in S}[\hat{F}_j(x)]_+$	**BNN (Brown–von Neumann–Nash)** Switches to strategies with positive excess payoff; innovations enter.
$\rho_{ij} = [\pi_j - \pi_i]_+$	$\dot{x}_i = \sum_{j\in S} x_j[F_i(x) - F_j(x)]_+ - x_i \sum_{j\in S}[F_j(x) - F_i(x)]_+$	**Smith** Pairwise comparison: mass flows in from worse-off strategies and flows out to better-off ones.

x_i — the number of players playing strategy i

$\pi_i = F_i(x)$ is the payoff to strategy i with population distribution x

$$\hat{F}_i(x) = F_i(x) - \sum_k x_k F_k(x)$$

is the excess payoff over average

$M(\pi)$ is the best response correspondence

9 Rock–Paper–Scissors

To examine how each dynamic works, consider Rock–Paper–Scissors. In this example, we assume that the population is divided into three groups corresponding to Rock, Paper, and Scissors, and that the interaction between any two individuals is governed by the payoff matrix:

	R (Rock)	**P** (Paper)	**S** (Scissors)
R (Rock)	0	-1	1
P (Paper)	1	0	-1
S (Scissors)	-1	1	0

Note that this is a non-zero sum game, so that the column player's payoff matrix is equal to negative one times this payoff matrix.

We can represent the population state as

$$x = (x_R, x_P, x_S),$$

where

$$x_R + x_P + x_S = 1.$$

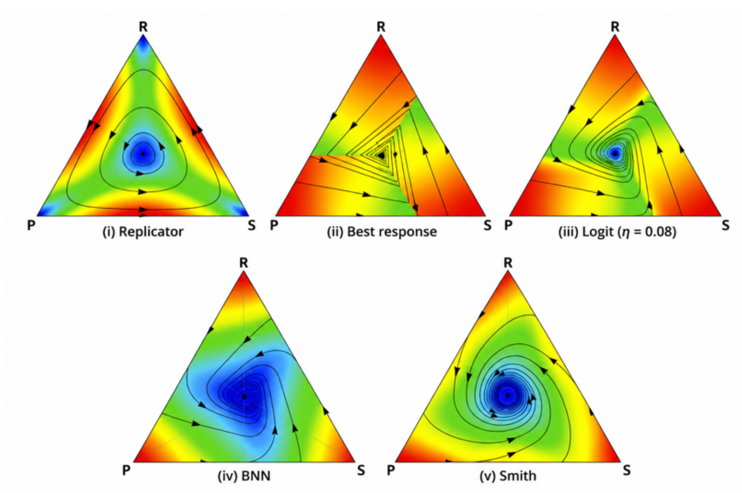
Thus, all solutions live on the simplex

$$\left\{x \in \mathbb{R}^3 : x_R, x_P, x_S \geq 0, \quad x_R + x_P + x_S = 1\right\}.$$

The unique mixed Nash equilibrium is

$$x^* = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

The population trajectory for each revision protocol was modeled by Sandholm in his *Population Games and Evolutionary Dynamics* textbook:



10 Evolutionary Dynamics and Mean Field Games

Recent research has explored connections between mean field games and evolutionary dynamics, as both study the evolution of distributions over large populations.

Both frameworks involve forward equations that describe how a population distribution evolves over time. In finite-state mean field games, the population state resembles the strategy distribution in a population game.

11 Open Questions

The main open questions from this week's investigation are:

1. Can we design a smooth activation function that preserves Nash equilibria as fixed points?
2. Does convexity of the activation function affect convergence?
3. Can Nash equilibria be recovered from time-averaged trajectories when they are not fixed points?
4. How does payoff scaling affect RK4 stability and convergence time?
5. Can BNN-style evolutionary dynamics be connected to finite-state mean field games?